English for Mathematics 1
Autumn term 2022

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September 20, 2022

Contents

Chapter 1

Introduction

1.1 Class hours

MA33002	**MA33002EHIT**
week 1-4 (last class 16th Sept.)	week 1-4 (last class 16th Sept.)
Tuesday 15:45 - 17:30	Tuesday 18:30 - 20:15
Friday 13:45 - 15:30	Friday 18:30 - 20:15

1.2 Homework

- A larger homework will be given on the last week (Tuesday, 13th Sept.).
- Submission deadline is Tuesday, 20th Sept., 18:00. Submit by
  - email at f.bossmann@hit.edu.cn
    use your HIT student email address
    attach the homework file, don't use any online storage!
  - in the last class (16th Sept.)
  - at my office (Science Park, TIB K1344)
    20th Sept. , 13:00 - 18:00

  Don't forget to put your name, number, and the lecture number on the submission.
- **The homework is necessary to pass the class!** There is no exam.

1.3 Chapters of this class

Chapter 1: Introduction (this chapter)
formalities and introductory example

Chapter 2: Elementary building blocks of text
What are the components of a text?
How do we use them to structure our text correctly?

Chapter 3-7: From the book "The Elements of style"
by William Strunk Jr. and E.B. White (4th Edition)

- a collection of rules and guidelines for writing
- Grammar rules, stylistic rules, rules on information flow
  Do not break (without purpose)!
- The book is on general writing, the class is specificly for scientific writing
- From small to more complicated text structures (building blocks), but some rules can apply in different situations

Chapter 8: Linking words
A list of useful words to connect statements in various situations.

1.4 Why do I need to learn this?

Three reasons why a correct writing style is so important:

Grammar: Universal rules to ensure understanding!
Consider the equation $3 + 5 \times 2 - 1 = 12$. We all agree that the result is 12 and not 15 (left to right order) or 8 (addition/substraction before multiplication). You need to use brackets if you want to change the orders, as you have to use certain interpunctuation to change the meaning of a sentence.
Scientific publications address a wide audience from all around the world (native speakers from different countries, non-native speakers with varying English levels). Hence, it needs to be standardized to asure universal understanding (not like e.g., local dialects).

Readability: Text needs to be easily readable and understandable. It should not sound boring or monotone.

Today, there are thousands of new publications each month. The preprint platform ArXiv alone registers around 15.000 submission per month. It is hard to gain recognition among all those, especially as young unknown researcher. Moreover, it is impossible for other researchers to fully read all new publications in their field.

The text needs to be written in a way, that it keeps the readers attention. It must not be too complicated to understand, too monoton, or too boring. Most important part are title and abstract.

Information flow: Text is basically an encoding of information. You need to choose your formulations carefully to asure the reader can extract all information in exactly the way you intended it.

Most researchers will not read all publications completely from start to end. This would be way too time consuming. Instead they briefly skim over the text to decide if the work is worth reading in detail. Hence, you need to place the most important information just at the right places to not be overseen.

Also, different formulations can emphasize the information in different ways. Compare the following three examples:

Our method failed in 5% of the test cases.
Our method failed in only 5% of the test cases.
Our method was successful in 95% of the test cases.

Note that these reasons hold independent of the language. They also apply in parts to presentations and talks. Many of the rules we will learn can also be applied to other languages or spoken text.

1.5 An introductory example

Example 1:

In figure 7 the three obtained data is shown. We compared our method (marked in red colour), to earlier technique (marked in blue colour) and - as you can clearly see - it does not favor the latter one.

What are the grammatical mistakes in this text?

- Figure instead of figure (”Figure 7” is considered a name)
- data is an uncountable noun, so ”three data” is not possible. (You can e.g., use ”three data points” or ”three data sets” instead. Then, the sentence needs to use the plural form ”are shown”.)
- the comma ”...red colour),” is wrong
- ”earlier technique” should be ”an earlier technique” or ”earlier techniques”
- ”colour” is British English while ”favor” is American English

Example 2:

The obtained data is shown in Figure 7. We compared our method (marked in red color) to earlier techniques (marked in blue color) and - as you can clearly see - it does not favor the latter one.

Can we improve readability on this text?

- Remove passive voice when possible ”obtained data is shown”
- Remove unnecessary words (”marked in red colour” to just ”red”)
- ”Compared to” should be used for similarities, ”Compared with” for differences
- Split sentences into two, if they are too long
- Avoid dash
- Instead of ”You” better use ”We” or ”The reader”
- ”it” is ambiguous
- Avoid negative language
- Emphasize on your results

Example 3:

Figure 7 shows the obtained data. We compared our method (red) with earlier techniques (blue). As we can clearly see, the obtained data favors our method.

Chapter 2

Elementary building blocks of text

We use text to encode our thoughts, conclusions, and ideas. In the process of writing, we often face the difficulty that those ideas can form very complex structures that are hard to describe. Especially in scientific writing we have to deal with one restriction of text: it is linear, always read from start to end.
A fictional story is often told in chronological order (unless for suspense or if time travel is involved). Science however is, in many cases, non-linear:

- Your work is always based on several previous works (or parts) that you will refer to:

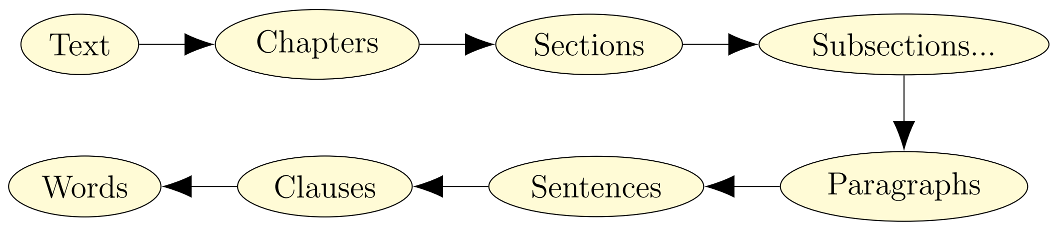
  For the proof we need Theorem 10 in the work of Donoho [1].
  Smith et. al. developed a similar algorithm (see publication [10]).

- Within your work you will often refer back to previous sections for your arguments.

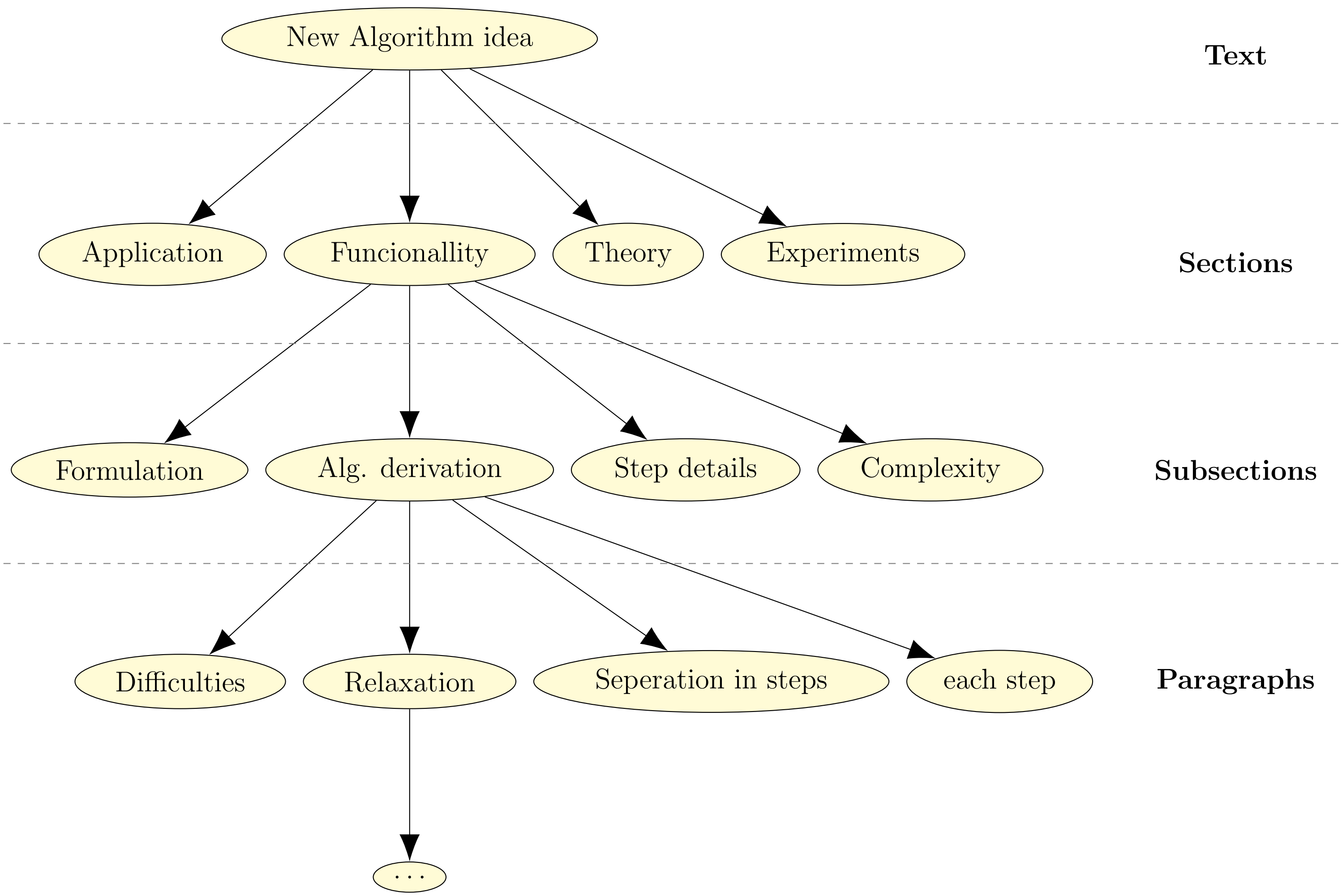
  By equation (3) it follows that the system matrix is sparse.
  Note that Theorem 11 is a generalization of Theorem 2.
  For the applications listed in Section 2, this constant is small.

It is simply not possible to avoid these references to other works or earlier sections. Hence, it is most important that we understand how a text can be structured to help overcoming this restriction. A neat structured text also increases readability and is often much easier to understand.

What are the building blocks of a text? (large to small)



Not all blocks are used in every work. Normal journal papers, for example, do not have chapters but start with sections as blocks. More complex works, such as a PhD thesis, can have sub-subsections or even subsub-subsections. As a general rule, each block should not be too short or too long. One building block should never contain only one other block (e.g., a paragraph should never be only one sentence or a section never only one paragraph).
The rules we learn in later chapters will help dividing the text into meaningful blocks. To see, how this can help to transfer our ideas into text, we consider the following example: Assume we want to publish a new algorithm that we developed. The algorithm solves a given problem by dividing it into multiple steps. We would like to publish this in a scientific journal.


New Algorithm idea
Text
Application
Funcionallity
Theory
Experiments
Sections
Formulation
Alg. derivation
Step details
Complexity
Subsections
Difficulties
Relaxation
Seperation in steps
each step
Paragraphs
...

Chapter 3

Elementary rules of usage

This chapter (and also the following chapters) are based on the book "Elements of Style". The rule numbering is the same as in the book, but the examples as well as some rules have been adjusted to better fit scientific (mathematical) writing. This chapter introduces eleven basic rules of writing. We cover important grammar rules and correct punctuation. The rules apply to the building blocks words, clauses, and sentences.

Rule 1:

Form the possessive singular of nouns by adding 's.

"Possesive singular" is the case of a noun, that indicates ownership. It means that something belongs/is owned by the noun.

The phone of the teacher. The teacher's phone.

The ownership is not restricted to physical objects, but can also be abstract.

The last theorem of Fermat. Fermat's last theorem.
The thesis of Charles. Charles's thesis.

Sometimes you see the "s" omitted when the noun already ends on "s" itself (*Charles' thesis*). Both forms are correct and it rather depends on which style guide one follows. If you are unsure about the style, use an alternative formulation.
This rule also applies for indefinite pronouns. Possessive pronouns on the other hand form their own words. (Pronouns are words that can be used

instead of a noun - replacing it. Indefinite pronouns do not specify what exactly they refer to. They can be used if the owner is unknown or not important.)

> *One's reference, somebody else's choice.*
> *This is my (his, her, their, your) book.*
> *This book is mine (his, hers, theirs, yours).*

A common mistake is to confuse the possessive pronoun "its" with the short form "it's" for "it is".

> *It's surprising how good this AI has become. Its results in last years are amazing.*

Rule 2:

> *In a series of three or more terms with a single conjunction, use a comma after each term except the last one.*

A conjunction is a connecting word, joining together sentences, clauses, or words. This rule is also known as the Oxford comma. We place a comma after each element of a list. This includes a comma before the last term (before the conjunction!).

> ***Wrong:*** *Let f be a continuous, bounded and convex function.*
> ***Right:*** *Let f be a continuous, bounded, and convex function.*

The Oxford comma was introduced to remove ambiguities and avoid confusion.

> ***Wrong:*** *We discuss several cases: convex, concave and bounded.*
> ***Right:*** *We discuss several cases: convex, concave, and bounded.*

Without the comma, it is hard to tell if there are two or three different cases. Is there one case "concave and bounded" or two cases "concave" and "bounded"?

Rule 3:

Enclose parenthetic expressions between commas.

A parenthetic expression is something that is added to the original sentence without changing its meaning or grammar. It just gives additional information.

> *Note that our result, in contrast to previous works, also covers the complex case.*
> *The constant C, which is independend of ε, can be bounded from above.*

Never only use one comma when the phrase is in the middle of the sentence! Two commas separate the expression from the rest of the sentence and make it clear and easy to read. Only using one of the commas can lead to big confusions. You may use no comma at all when the phrase is only one word (but be consistant throughout your work!).

> *However, this will be subject to future works.*
> *However this will be subject to future works.*

> *Although f is non-convex, we can, nevertheless, calculate its global minimum.*
> *Although f is non-convex, we can nevertheless calculate its global minimum.*

Note that there is a comma before abbreviations (i.e., e.g., etc.). In American English there is also a comma afterwards, not so in British English. Most journals require American English. The abbreviation "i.e." stands for "id est" (latin) and translates to "this means" or "that is"; "e.g." stands for "exempli gratia" (latin) and translates to "for example". Both abbreviations are commonly used in scientific writing.

> ***AE:*** *The function is convex and bounded, i.e., its minimum exists.*
> ***BE:*** *The function is convex and bounded, i.e. its minimum exists.*

Remember, parenthetic expression give additional information. The are non-restrictive. No comma should be used if the information is a mandatory explanation or restrictive term.

> ***Additional:*** *We can prove this result for the given function f, which is convex.*
> ***Mandatory:****We can prove this result for functions f that are convex.*

As a simple check, remove the expression and check if the sentence can still be understood and if the meaning is the same.

> *We can prove this result for the given function f.*
> *We can prove this result for functions f.*

Although both sentences are grammatically correct, the meaning of the second example now changed. Suddenly, we are able to prove the result for all functions and not only for convex ones.

Rule 4:

> *Place a comma before a conjunction introducing an independent clause.*

An independent clause is a clause that clause that expresses a complete thought or statement. It is usually a sentence in its own.

> *Here, x can be chosen arbitrarily, but y must be rational.*
> *Here, x must be an integer, and y must also be rational.*

Note that both examples could easily be split into two sentences that are not connected. This normally indicates that we have an independent clause. With the right choice of the conjunction we can form a connection between both clauses. In the first example we use "but" to emphasize that the two statements are contrary (free choice of x / restriction on y). Other conjunctions are

> *and, but, as, for, while, (either...) or, neither... nor, ...*

The only exception is, if both clauses share the same subject and are connected by "and". This combination indicates a strong relation between both clauses. Hence, the comma can be omitted.

> *Here, x must be an integer and smaller than 100.*
> *Here, f must be bounded from below, but can tend to* $+\infty$.

Rule 5:

Do not join independent clauses with (only) a comma.

Never join two (or more) independent clauses using only a comma and no conjunction. If we don't use a conjunction, then we have to separate the clauses using a semicolon ";" or a period / full stop "." . A comma is a too weak separation to expresse the independence of the clauses.
When deciding how to connect two clauses, use the following rule of thumb. A conjunction with comma expresses a rated connection. A semicolon is more uncommen and presents a connection similar to "and". A period is easier to read and should be used if the clauses are long or more complex.

> *Here, y must be rational, while x can be chosen arbitrarily.*
> *Here, x can be chosen arbitrarily; y must be a rational number.*
> *Here, x can be chosen arbitrarily. The variable y must be rational, positive, and smaller than $x^2 + 13$.*

There is one exception from this rule. If there are multiple very short and similar statements. In this case we can join then in a list and Rule 3 (Oxford comma) applies.

> *Here, x must be real, y must be rational, and z must be positive.*

Rule 6:

Do not break sentences in two.

Do not use periods for commas. Do not separate clauses that are connected or depend on each other (see previous rules). If you want to form two sentences, because e.g., the clauses are quite long and complex, you need to be careful. Starting a sentence with a conjunction is grammatically not wrong, but considered bad writing in many cases. If both clauses share a noun you can use demonstrative pronouns (this, that, these, those,...) to link them.

> ***Wrong:*** *Here y must be rational. While x can be chosen arbitrarily.*
> ***Right:*** *Here y must be rational. The variable x however can be chosen arbitrarily.*

> ***Wrong:*** *The algorithm is known as Newton's method. A technique to find a local minimum of a function.*
> ***Right:*** *The algorithm is known as Newton's method, a technique to find a local minimum of a function.*
> ***Right:*** *The algorithm is known as Newton's method. This is a technique to find a local minimum of a function.*

Rule 7:

> *Use a colon after an independent clause to introduce a list of particulars.*

A colon ":" can be used to introduce a list of terms. The clause before the colon needs to be independent(!). This means, it needs to form a sentence on its own. Typical mistakes are, to separate a verb from its complement (a phrase that follows the verb to complete its meaning), or a preposition (in, at, on, such as,...) from its object.

> ***Wrong:*** *There are several publications on this topic such as: [3] for convex functions, [4] for Lipschitz continuous functions, and [5] for bounded functions.*
> ***Right:*** *There are several publications on this topic: [3] for convex functions, [4] for Lipschitz continuous functions, and [5] for bounded functions.*

Here the numbers [3], [4], and [5] refer to the third, fourth, and fifth entry in a list of publications that is usually given at the end of a scientific work.

> ***Wrong:*** *We discuss different cases where f is: convex, concave, and bounded.*
> ***Right:*** *We discuss three different cases of f: convex, concave, and bounded.*

Rule 8:

> *Avoid using dash.*

The use of a dash "-" is quite uncommon in scientific writing. Avoid using it and instead reformulate the sentence. Often there are more suitable formulations.
The dash sets of an abrupt break that introduces a summary or a long appositive (further description of some statement).

Wrong: *We use gradient descent - a technique similar to Newton's method but without requiring the second derivative - to solve the problem.*
Right: *We use gradient descent to solve the problem. This technique is similar to Newton's method but does not require a second derivative.*

Rule 9:

The number of the subject determines the number of the verb.

This rule sounds simple but has many pitfalls. If there is only one singular subject, then the verb should also use the singular form. For more subjects or a subject in plural form the verb should be plural. However, sometimes it can be tricky to find the number of the subject.

The function f is convex.
The functions f, g, and h are convex.
The set S, which includes the functions f, g, and h, is convex.
The set S and the functions f are convex.

Especially for pronouns, deciding wether to use singular or plural can be tricky. After **each, either, every, neither (as conjunction), no, one,** ... the singular form is used. These pronouns refer to a single element of a group or to one group as a unit.

Each of these functions is convex.
There exists no positive integer smaller 0.
Five and seven are both positive integer, but neither is even.

Plural is used after pronouns that refer to multiple elements, such as **some, all,**.... If "neither" negates two or more things collectively, it is also sometimes followed by the plural form (some style guides may consider this too informal). This is often the case when it is used as **neither of**.

Some of these functions are convex.
All of these functions are convex.
Neither of these functions are convex.

The pronouns "any" and "none" can be followed by both singular and plural form. If "any" is followed by a countable noun, then the number of the noun defines the number of the verb. For uncountable nouns, the singular form of the verb should be used (see later example). The pronoun "none" can be used as "not one" or "not any". Hence, the singular or plural form is defined as for "one" or "any" depending on which case is used.

Is there any convex function in the set? (find one)
Are there any convex functions in the set? (one or more)

None of these functions is convex. (not one)
None of these functions are convex. (not any)

If there are two or more subjects, normally the plural form is used. However, certain compounds can form a unit and thus the singular form is used.

Newton's method and gradient descent are used to find minima.
Divide and Conquer is a good strategy in this case.

Sometimes there are several nouns in the sentences. The form of the verb always depends on the subject.

The difficulty of these problems is the large number of variables.
These problems have a large number of variables.
The large number of variables is a difficulty of these problems.

For uncountable nouns, such as information or data, always use the singular form of the verb. You can recognize these nouns by trying to count them (whithout the help of other words!). What is two information, or five data? Some uncountable nouns may look like a plural form (e.g., Mathematics), but they still go with a singular verb.

Information was gathered during the experiment.
The data is provided by our research partner.
Mathematics is a beautiful subject.

Never use "a" or "an" with uncountable nouns (an information, a data). The noun "data" is sometimes also used as plural form of "datum" and then followed by the plural form of the verb. This use is not so common and mostly limited to British English. If you are unsure whether to use singular or plural, you can force one or the other by using auxiliary words.

This piece of information was gathered during the experiments.
Those data points are provided by our research partner.

Rule 10:

Use the proper case of pronoun.

A pronoun (I, you, he, she, it,...) are declined depending on their use in the sentence. There are three different cases a pronoun can take: subject of the sentence (I, she, they), object of the sentence (me, her, them), showing possession of something (mine, hers, theirs). If the pronoun is subject and object at the same time, the reflexive form can be used as object (myself, herself, themselves). Possession can also be shown using possessive adjectives (my, her, their).

We developed this algorithm. (subject)
The algorithm was developed by us. (object)
This algorithm is ours. (possessive)
We developed this algorithm by ourselves. (reflexive)
This is our algorithm. (possessive adjective)

The pronoun "who" can be used when formulation questions, or starting a relative clause (a clause adding or modifying information). It is declined as who (subject), whom (object), and whose (possessive). However, the object form "whom" is more and more replaced by "who" especially in informal language.

We are the researchers who developed this algorithm. (subject)
By whom was this algorithm developed? (object)
We contacted the author whose algorithm we use. (possessive)

Sometimes it is possible to omit some words when using pronouns in comparisons. However, it can lead to misunderstandings and should be avoided.

We present a more efficient algorithm than they (do).
We present a more efficient algorithm than theirs (their algorithm).

Rule 11:

A participle phrase at the beginning of a sentence must refer to the grammatical subject.

A participle is a verb that is used as adjective ("the hard working student passed the exam without problems"). A participle phrase is a phrase containing a participle. In general, the phrase can be used as adjective on any noun in the sentence. Is the participle phrase placed at the beginning of the sentence, then it needs to refer to the subject of the sentence. You also need to separate it with a comma.

> *Assuming convexity, our result coincides with the work [3].*
> *Assuming convexity, the work [3] coincides with our result.*

In the first sentence, our result is more general and [3] only covers the convex case. In the second sentence it is exactly reversed.

> ***Wrong:*** *Exponentially decreasing in absolute value, we conclude that the function converges to 0.*
> ***Right:*** *Exponentially decreasing in absolute value, the function converges to 0.*

In the first case, we are the ones who are exponentially decreasing. If a participle phrase is not used at the beginning of a sentence, then it should be set as close to the noun it modifies as possible. If it contains non-essential information then it needs to be separated with commas.

> *We conclude that this function, exponentially decreasing in absolute value, converges to 0.*

Note that the information given here by the participle phrase is not essential for the sentence (in a grammatical sense!). It is already clear that we are talking about "this function". (The information provided might be essential for the mathematical argument though.)

> *For this reason, we use the function having the fastest decay rate in our experiments.*

Here the participle phrase is essential to identify the correct function (out of a given set of functions). "We use the function which has the fastest decay rate in our experiments."

3.1 Exercise

The following text violates each of the eleven rules. Mark all mistakes and rewrite the text in a correct form.

Example 4:

In [3] the authors introduces a new algorithm to calculate the minimum of a function. The method is similar to Newtons method but it avoids calculating the second derivative. They proposed method can be applied to: convex, bounded and quadratic functions. In many applications such as machine learning or image denoising the new method - compared to older techniques - shows remarkable results. We will use this algorithm for video denoising. Which is a three dimensional problem. This should improve the results drastically, the runtime, being iterative, of the algorithm can unfortunately be quite long.

Chapter 4

Elementary Principles of Composition

This chapter contains eleven rules covering the building blocks sentences, paragraphs, and sections. All rules are about the writing style and not about grammar. It is adviced to follow these rules. However, they can be broken given a valid reason, e.g., to break up an otherwise to monoton text.

Rule 12:

Choose a suitable design and hold to it.

This rule sounds most general at first and it may not seem clear at first what exactly it means. Nevertheless, it might be one of the most important rules of this lecture. Choose a suitable design **in the beginning** of the writing process. Force yourself to stick to this design. This will lead to a clearly structured work. Before you start writing, ask yourself

- Which results of my work do I want to cover, and which can be omitted?
- What are the most important results?
- Which chapters / sections do I need?
- What is the best order to present my work?
- What variables and notation do I need?

or more detailed questions like

- Do I introduce all notation in the beginning or each part just when its needed?

- Do I need an Appendix e.g., for proofs?
- Do I use definition, theorem, proposition, lemma, corollary, or other building blocks? If yes, which part of my result belong in which block?
- Which part of my work do I need to refer to later in the text? These parts need to be easy to find, best numbered.

Answering all these questions beforehand will help you getting a clear image of your work. Writing will become much easier afterwards. If you have trouble with some of the questions, you can always check which style and notation related works use. It is good advise to stick to the commonly used forms of your research field. If you are publishing in a scientific journal, also check the "author guidelines" for the required standards.
During the writing process it might happen, that you want to break your design at some points. Ask yourself if this is really necessary or if you can avoid it. Should it happen too often, you might have to overthink your design choice. Be careful, if you change your design afterwards, you need to check the previously written text for compability with the new design.

Rule 13:

Make the paragraph the unit of composition.

The paragraph should form the smallest self-contained unit of your work. It normally holds exactly one thought / idea / case / step of your work. Connected ideas are then grouped in (sub)section which form the next higher building block, and so on. Each sentence of the paragraph itself contributes in explaining the idea, but is no longer self-contained. Taking it out of context will not provide any useful information.

Example 5:

Let us now discuss the case where f is convex. By Theorem 10 it follows that the function has (at least) one global minimum. Using equation (10) we can also see that this minimum must be unique. We can simply apply Newton's method to find it. Hence, the convex case is solved.

Consider the above example. The paragraph contains exactly one idea, namely how to find the minimum of f when f is convex. Take each sentence on its own and the information is lost. Note that the first sentence introduces the paragraph's idea and also links it to the previous one ("now" implies that the paragraph before most likely discussed a different case, e.g., where f is concave). Other formulations might be

> *We can prove the convex case using similar arguments.*
> *Let us first discuss the convex case.*
> *For the convex case we need to follow a different approach.*

Equally, the last sentence gives a short summary of what was achieved in this paragraph. This structure increases readability and comprehension especially for longer paragraphs. There are some rules that should always hold for paragraphs.

- It should hold exactly one thought / idea . . ..
- A single sentence should never form a paragraph.
- Paragraphs shouldn't be too long (wall of text).
- Too many short paragraphs in a row look chaotic.

Rule 14:

Use the active voice.

This is the first of several rules, that will increase the quality of your work in many ways. The active voice is often, shorter, easier to read, and more clear in its meaning than the passive voice. It also can sound much stronger.

> ***Wrong:*** *It will be shown that other techniques are outperformed by our algorithm.*
> ***Right:*** *We show that our algorithm outperforms other techniques.*

Note how the connotation of both sentences is largely different. The second sentence has a much higher focus on our work and algorithm. Sometimes the difference is more subtle and the right choice depends on the context.

> *Convergence can be proven by applying Theorem 10.*
> *We prove convergence by applying Theorem 10.*
> *Convergence follows by applying Theorem 10.*
> *We apply Theorem 10 to prove convergence.*

In some cases the difference is so small that both forms can be used, depending on what fits best in the text:

> *The obtained results are shown in Figure 3.*
> *Figure 3 shows the obtained results.*

Rule 15:

Put statements in positive form.

Make rare use of negative forms (not, nor, unable, ...) and conditionals (would, should, could, may, might, ...). Negative statements suggest the reader that there is missing information or unfinished work. Some worlds like "only" can have a similar effect. Conditonals sound full of doubt and present a lack of authority.

> ***Wrong:*** *In this work, we will not consider the concave case.*
> ***Better:*** *In this work, we will only consider the convex case.*
> ***Right:*** *In this work, we will concentrate on the convex case.*

To strenghten the statement even more, you can follow it with a sentence like these.

> *We will address the other cases in future works.*
> *This is the most important case for our application.*

Here is an example showing how doubtful conditionals can sound.

> ***Wrong:*** *If the function f would be convex, we could find its global minimum. However, since f is non-convex, we may only find a local minimum.*
> ***Right:*** *In the case where f is convex, we can easily find the global minimum. However, for non-convex f we can still find a local minimum.*

The effect of negative forms can sometimes be used to especially emphasizes a positive statement.

> *Not only does our algorithm converge, but it also does so in only a few iterations.*
> *The precision of our algorithm cannot be achieved with other methods.*

Rule 16:

Use definite, specific, and concrete language.

You do not need to give all details, but always state the important facts that are needed to understand the very next steps. Use concrete and clear language. This can be

- written text,
- mathematical formula,
- explaining the principle on simple examples,
- using figures and graphs,
- giving applications for an abstract concept,
- restricting yourself to an easy to understand special case first,
- or a combination of the above.

> ***Wrong:*** *We use some properties of f to find its global minimum.*
> ***Wrong:*** *The function f is convex, bounded, real-valued, smooth, and symmetric. Hence, we can find its global minimum.*
> ***Right:*** *We use the convexity of f to find its global minimum.*

Sometimes using only text is not sufficient to explain a concept.

> *The convex conjugate $f^*(y)$ of a function $f(x)$ is defined as the supremum of the difference of the inner product $\langle x, y\rangle$ and the function itself.*

We can use a combination of other forms:

> ***Mathematical formula:*** *The convex conjugate f^* of a function f is defined as*
>
> $$f^*(y) = \sup_x \langle x, y\rangle - f(x).$$
>
> ***Example:*** *Exemplary, let $f\mathbb{R} \to \mathbb{R}$ be $f(x) = |x|$. Then*
>
> $$f^*(y) = \sup_x xy - |x| = \sup_x |x|(\operatorname{sign}(x)y - 1) = \begin{cases} 0 & |y| \leq 1 \\ \infty & |y| > 1 \end{cases}$$
>
> ***Image:*** *Figure 1 shows the graph of this function.*

Sometimes the smallest sentences can contain uncertainties.

> *Let $x \in \mathbb{N}$. (Is 0 included or not?)*
> *Let $x \in \mathbb{N}_0$.*
> *Let $x \in \mathbb{N}$, where $\mathbb{N}$ does (not) include 0.*

Rule 17:

Omit needless words.

Do not use filling words that don't carry any information. Ask yourself for every word: What information does it contain and is it really necessary? Often sentences can be formulated more short and compact.

> ***Wrong:*** *We use the fact that f is convex to find its minimum.*
> ***Right:*** *We use the convexity of f to find its minimum.*

> ***Wrong:*** *This is a subject that was already discussed in former works.*
> ***Right:*** *This subject was already discussed in former works.*
> *This is a subject already discussed in former works.*

Depending on the context and desired connotation, the words "subject" and "already" can also be omitted: "This was discussed in former works.".

> ***Wrong:*** *For the sake of simplicity, we use the short notation.*
> ***Right:*** *For simplicity, we use the short notation.*

Note that some word do not directly contain information, but are important for for the connotation or emphasis.

> *Our algorithm still returns reasonable results, even when the data is highly noised.*
> *Our algorithm outperforms the other techniques, especially in the convex case.*

Rule 18:

Avoid a succession of loose sentences.

A succession of loose sentences is a number of sentences that are not connected in any way. Meaning, they do not link to each other and they do not use conjunctions to connect. This can appear boring and monoton.

> ***Wrong:*** *Let f be convex. Consider x as integer. Choose y such that xy is smaller 1. ...*
> ***Right:*** *Let f be convex and x be any integer. Furthermore, choose y such that xy is smaller 1. ...*

The effect can be even more extrem when all sentences consist of two clauses with "and".

> ***Wrong:*** *Let f be convex and x be an integer. Choose y such that xy is smaller 1 and z such that xz is larger 1. Let f be differentiable and apply Theorem 10. ...*
> ***Right:*** *Let f be convex and differentiable. Furthermore, for an integer x, choose y and z such that xy is smaller 1 and xz is larger 1. Now apply Theorem 10.*

Note how we use different sentence structures to make the reading less monoton. We also use linking words to form a connection between the sentences.

Rule 19:

Express coordinate ideas in similar form.

We can show the similarity of two ideas by using the or very similar constructs in text to describe them.

> *For any convex function on a bounded interval, the global minimum exists.*
>
> ***Wrong:*** *The global maximum exists, if we have a concave and bounded function.*
> ***Right:*** *For any concave function on a bounded interval, the global maximum exists.*

Note how we only changed two words in the correct formulation to express that both statements (convex and concave case) are very similar. This rule also applies to mathematical statements in the wider sence. If you have two Theorems with similar results, use the same notation and similar formulations.
If you are using a list with prepositions (in, on, by, ...), then the prepositions goes either only to the first element or to all.

> ***Wrong:*** *The functions used in Theorem 10, Lemma 12, and in Proposition 14 are convex.*
> ***Right:*** *The functions used in Theorem 10, Lemma 12, and Proposition 14 are convex.*
> ***Right:*** *The functions used in Theorem 10, in 12, and in Proposition 14 are convex.*

Sometimes the preposition is not the same for each entry. Then the correct one needs to go to each element.

> ***Wrong:*** *We prove this by substituting equation (14) and applying Theorem 10 to equation (16).*
> ***Right:*** *We prove this by substituting equation (14) into and applyint Theorem 10 to equation (16).*
>
> *We prove this by substituting equation (14) into euqation (16). Then we can apply Theorem 10.*

When using correlatives (both, and, not, not only, ...) to express similarities, then each part should follow the same construct.

> ***Wrong:*** *We prove not only convexity but also show that the set is bounded.*
> ***Right:*** *We show that the set is not only convex but also bounded.*
>
> ***Wrong:*** *The set is both, convex and it can be bounded.*
> ***Right:*** *The set is both, convex and bounded.*

On first sight it seems that Rule 18 and Rule 19 are contrary to each other. Rule 18 forbids to use the same structure over and over, while Rule 19 forces the same structure for similar statements. However there two points to keep in mind.

- Rule 19 does not only apply for statements following directly after another. They can be in completely different parts of the document. Consider two similar theorems. Theorem 3 for the one dimensional case very early in the document, and Theorem 40 for the multi-dimensional case much later. They should still be formulated similarly. Moreover, Theorem 40 should be followed by a remark "see also Theorem 3 on Page 12". This allows the reader to easily compare both results.
- If more then a few statements are similar to each other and applying Rule 19 would result in a monoton text, the statements can be split into smaller groups. Alternatively, a table or list can be used to express the similarities.

Rule 20:

Keep related words together.

It is natural, that you want to avoid using the same formulations over and over again. At some point, everyone starts experimenting with the order of words to create new and more fancy sentences. While they still might be grammatically correct, they are often harder to read and understand. When forming a sentence, always consider which words are connected and keep those words close to each other.

> ***Wrong:*** *The function f, which was introduced in [3], is convex.*
> ***Right:*** *The convex function f was introduced in [3].*

> ***Wrong:*** *Note the artifact shown in Figure 3 that is right in the center.*
> ***Right:*** *Note the artifact right in the center of Figure 3.*

Never separate a subject and its verb by a phrase.

> ***Wrong:*** *The function f, convex and bounded, can be minimized.*
> ***Right:*** *The convex and bounded function f can be minimized.*

> ***Wrong:*** *We can, similary to the previous chapter, prove that f is convex.*
> ***Right:*** *Similar to the previous chapter, we can prove that f is convex.*

Relative pronouns should come immediately after its refered noun.

> ***Wrong:*** *We use this function later on, which is convex.*
> ***Right:*** *We use this function, which is convex, later on.*

Sometimes this can cause ambiguities. In this case you can change the position or reformulate the sentence.

> ***Ambiguous:*** *The derivative of the function, which is convex, is given in equation (15).*
> *The convex derivative of the function is given in (15).*
> *The derivative of the convex function is given in (15).*

Modifiers should go next to the words they modify.

> *All the functions are not convex.*
> *Not all the functions are convex.*

Rule 21:

In summaries, keep to one tense.

Summaries of your work, such as the abstract (at the beginning of a work) and the conclusion (at the end) should always be writtne in one tense. Usually the present tense (We present, the results are, ...) for the abstract and past tense (We presented, the results were, ...) for the conclusion.
The usual tense throughout the complete work, with exception of the conclusion, should be present tense.

We introduce a new algorithm.
We demonstrate its convergence in numerical examples.
Let f be a convex function.

Some sentences that seem to require past tense, can be written reformulated into present tense.

Past: *We plotted the data in Figure 3.*
Present: *The data is plotted in Figure 3.*
Present: *Figure 3 shows the data.*

Past: *The numerical setup was as follows.*
Present: *The numerical setup is as follows.*

Past: *Former works were published on this topic.*
Present: *Former works are published on this topic.*

Here "we plotted", "the setup was", and "works were published" all refer to past events and thus use past tense. We can replace it by events that are still present now. "The data is plotted (and still visible now) in Figure 3." Instead of "(our) setup was as follows" we use "(if you want to reproduce the results, then) the setup is as follows". And last, "were published" refers to the process of publishing (submitting the work, review process, etc.,) while "are published" refers to the state of being publicly availabe (right now).
There are only few exceptions where past tense can be used. First, in the conclusion where we summarize the work that lies now in the past from a readers perspective. The reader finished reading and now is looking at the conclusion.

We introduced a new algorithm. Numerical results demonstrated its great performance.

Second, if a specific event or date in the past is addressed. However, remember to only include necessary information and omit needless words. Is this specific event or date really of relevance?

> *The author published this result in 1978, twenty years before any other known results.*

Rule 22:

> *Place the emphatic words of a sentence at the end.*

The most prominent position of a sentence is at the end. The words at the end of a sentence are the ones getting the most attention by the reader. Hence, the information carried by these words is more likely to be picked up and perceived as more relevant.

> *The algorithm shows linear convergence.*
> *The algorithm converges linearly.*

The first example emphasizes that the algorithm is converging, while the second example focuses on the linear convergence rate. Make sure to put the most relevant information last.

> ***Wrong:*** *The problem is discussed by the author [2] as well as by others [3,4].*
> ***Right:*** *The problem is discussed by others [3,4] as well as by the author [2].*

The second most prominent position is the beginning of the sentence (if it doesn't start with the subject).

> ***Not prominent:*** *The algorithm shows linear convergence.*
> ***Prominent:*** *Unlike other methods, our algorithm shows linear convergence.*

This rule does not only apply to sentences. It also holds for paragraphs, (sub)sections, chapters, and other building blocks. This means, the last sentence of a paragraph, least last paragraph of a section, and the last section of a chapter are the most prominent ones. The second prominent position is the first sentence, the first paragraph, and the first section.

You can make a self experiment. Try getting a rough overview of an unknown writing as fast as possible. At some point you will start skipping words, sentences, and even complete paragraphs. You will read the first few words/sentences, then pick some information randomly from the middle part, and finally read the end. However, no matter how much text you skip, you will nearly always read the start and the end of a sentence / paragraph / section.

4.1 Exercise

Choose on of the rules of this chapter and write a text that violates this rule. We will collect all examples in the class and discuss some randomly chosen ones.

Chapter 5

A few Matters of Form

This chapter of the book covers topics such as headings, margins, numerals, quotations, and references. If you are interested in these topics for non-scientific writing, you can check out the chapter in the book.
For scientific writing, however, these things are usually already fixed by the requirements of the journal or university you want to publish. Hence, we do not have any freedom of choice here. Journals normally provide a "guideline for authors" that lists all their style requirements, along with templates for LaTeX or Word that fulfill these. We will learn more about authors guidelines, LaTeX, and the complete publication process in the Master lecture "English for Mathematics 2".
For the sake of this lecture, there are only three rules we need to learn.

Rule 23:

Do not use slang words and exclamation marks.

Slang words should not be used in any scientific writing. They might not be understood by everyone and also come with a special kind of emphasis. Do not use sentences like

> *The results were epic!*
> *Not a chance that other methods would beat us!*

Exclamation marks are used to put extra emphasis on the sentence. They usually indicate very strong feelings. As scientific writing is more about facts than feelings, exclamation marks should not be used.

Rule 24:

When adding a clause in parantheses to a sentence, the punctuation goes after the closing bracket. Punctuation inside the paranthesis follows normal rules, but do not put a period before the closing bracket.

Parantheses are a good way to add information to a sentence. parantheses express that the included content is an additional or minor information, sometimes just a clarification or example. Do not include information that is necessary for the understanding in the parantheses.
The parentheses always go before the punctuation of the main sentence. Punctuation inside parentheses follows normal rules with one exception: If the included text would end on a period, it is omitted instead.

> *We have proven convergence (see Theorem 11) and error bounds (see Theorem 12) for the algorithm.*
> *The function is convex and bounded (We do not prove this here. However, it is easy to see from the definition).*
> *The function is convex (and bounded?).*

Rule 25:

If the parentheses stand alone, the period before the closing bracket is not(!) omitted.

By omitting the period in the previous rule, we avoid writing ".)." what is punctuation clutter. When the parentheses are not included in a sentence, this clutter cannot appear. Hence the period is not omitted.

> *The function is convex and bounded. (We do not prove this here. However, it is easy to see from the definition.)*

5.1 Exercise

Rewrite the follwing two examples without using slang words.

> *Once in a blue moon the algorithm fails to converge.*
> *We wrap up this section with the following Corollary.*

Consider the following sentence.

The convergence of our algorithm follows from equation (4).

Now add the following information using parentheses if fitting:

- We also need Theorem 10 to prove the convergence.
- For the multi-dimensional case we need equation (14) and Theorem 13.
- The convergence rate will be discussed in a later section.

Chapter 6

Words and Expressions commonly misused

In this chapter we list some of the most commonly made writing mistakes. It involves similar words that are often confused, expressions that are grammatically wrong, and common ambiguous formulations.

among vs between

Both words make a statement about a subset of elements. Use "between" when you have a (definite) list of elements and "among" when talking about a group or set. A discrete list typically mentions all of the included elements, while a group is mentioned as a union in the text, without specifying each element.

> ***List:*** *We can choose between convex, concave, or differentiable functions.*
> ***Group:*** *We show that there is at least one convex function among those function.*

and/or

Sometimes you see the short form "and/or" used as conjunction. This is informal language and should not be used. It can also be irritating for readers that have never seen the formulation before. Better rewrite the text.

> ***Wrong:*** *The function is convex and/or bounded.*
> ***Right:*** *The function is convex, or bounded, or both.*

as good or better than

The same holds for this short form to express that something as as good as something else, or even better. There are plenty of better formulations that can be used.

> ***Wrong:*** *Our algorithm is as good or better than other techniques.*
> ***Right:*** *Our algorithm is as good as other techniques, if not better.*
> *Our algorithm is at least as good as other techniques.*

compare to vs compare with

You can compare two results to each other, but you can also compare them with each other. The difference of both formulations is subtle and often overseen. "Compare to" is looking for similarities between both elements, while "compare with" is looking for differences.

> ***Similarities:*** *Comparing Theorem 10 to Theorem 11, we see that both cover the convex case.*
> ***Differences:*** *Comparing Theorem 10 with Theorem 11, we see that the latter result is more general.*

data

We already covered this word in a previous chapter. However, it is a quite common term in scientific writing and thus worth repeating here. It is an uncountable word that can appear as both singular and plural. However, the singular form is much more common in scientific writing, the plural form is mainly used in British English.

> *The data was gathered in Beijing. We also got addditional data that were gathered by our British colleagues.*

different than

This formulation is getting popular in American English. While it is actually considered grammatically wrong, it gets more and more accepted. The correct way would be, to use "different from" or "different to".

> ***Wrong:*** *Our result is different than previous works.*
> ***Right:*** *Our result is different from previous works.*

divided into vs composed of

Both formulations describe an element that can be separated into smaller parts. "Divided into" is used whenever these smaller parts form a reasonable element by themselfs. Otherwise use "composed of".

A section is divided into paragraphs.
A natural number can be divided into its prime factors.
An apple is composed of seeds, pulp, and skin.
A fraction is composed of a numerator and denominator.

Facts

Be careful when using the word "fact". If some statement is an obvious fact, there is no need to extra mention it. Often the word "fact" is used in a situation, where it is not directly obvious. In the best case, this is lazy writing because one wants to avoid bringing the argument. In the worst case, the presented statement is actually not a fact at all. Also avoid similar formulations such as "it is well known".

It is a fact that all linear functions are convex.
It is a well known fact that each convex function has a unique minimum.

However vs nevertheless

Formerly, there was a difference between these two words. "However" was used whenever the meaning was "in whatever way" or "to whatever extend", and "nevertheless" was used in all other cases. However, nowadays the words are considered interchangeable. The only difference remaining is, that "nevertheless" is considered more formal and more emphatic than "however".

The function is not convex. However, we can still find its minimum.
The function is not convex. Nevertheless, we can still find its minimum.

less vs fewer

Both words express that there is a smaller amount of something. Use "fewer" when talking about countable elements; use "less" for quantities.

If we use fewer iterations, the result is less accurate.

nor

The word "nor" can only be used within the phrase "neither... nor". It cannot replace "or" in negative expressions.

Wrong: *The function is not convex nor concave.*
Right: *The function is neither convex nor concave.*

than

The word "than" is used when comparing different statements. Those sentences can become quite long. Hence, often words are omitted to make the sentence more compact. Be careful not to create ambiguities.

Wrong: *The variable x is closer to* 0 *than y.*
Right: *The variable x is closer to* 0 *than y is.*
Right: *The variable x is closer to* 0 *than it is to y.*

that vs which

Both words are used to add a describing clause to a noun. The difference is that "that" is used whenever the clause is restrictive / defining / necessary. Use "which" for non-restrictive clauses. Does the clause only give additional information and the sentence, whithout the clause, still has the same meaning? Then the clause is non-restrictive. If removing the clause changes the meaning of the sentence, then it is restrictive or defining.

Restrictive: *Among the given variables, choose one that is smaller than 1.*
Non-restrictive: *Among the given variables, choose x, which is smaller than 1.*

6.1 Exercise

Fill in the gaps.

Example 6:

Comparing Lemma 1 _____ Lemma 2, we see that both cover the complex case.
We use the set _____ was defined in the last section.
The homework is _____ three exercises.
There is no suitable choice _____ the set of linear functions.
Compared _____ later results, the old methods look underwhelming.
This argument is the same _____ was given in the last chapter.
When the input is _____ noisy, we require _____ iterations.
This technique can be _____ several steps.
We have to balance the method _____ numerical stability and efficiency.
Figure 3 shows the data that _____ gathered during the experiments.

Chapter 7

An Approach to Style

(Skipped in this lecture, will be tought in the Master lecture "English for Mathematics 2".)

Chapter 8

Linking words

So far in this lecture, we learned that our text should not be monoton. Clauses and sentences should be linked to visualize connections between the different statements. However, we did not learn how this can be achieved.
The easiest way to connect phrases and sentences is, to use so-called linking words. There is a vast number of these words and we will only cover a part of it in this chapter. Which linking word you use depends on the connection you want to express. Should the statements be added together? Do you have sequence of statements? Is one statement the cause or effect of the other? Here is a list of linking words ordered by category.

Sequence (link a sequence of statements)
First, firstly, the first, the second, second, third, thirdly, next, finally, last, in addition, an additional, moreover, furthermore, also, one, another, in conclusion, to summarize

> *First, we show that the function is convex. Next, we prove that it is bounded. In addition to this, we give a concrete upper bound. Last, we establish the new method. To summarize, our algorithm is developed using a convex and bounded function.*

Addition (add one statement to another)
In addition, furthermore, also, as well as

> *Our method can be applied in image processing as well as in optimization.*

Cause (One statement is the cause of the other)

For, because (of), since, as, to cause

Sine the noise on the data is high, the algorithm does not find a good solution.

Effect (One statement is caused by the other)

So, as a result, as a consequence, therefore, thus, consequently, hence, to result from, due to, of, to result in, to affect

The given data is highly noised. Consequently, the results are not good.

Emphasis (emphasize a statement)

Undoubtedly, indeed, obviously, generally, admittedly, in theory, in fact, particularly, especially, clearly

The results clearly show, that our method performs much better.

Comparison (set two statements in comparison)

Similarly, likewise, also, too, (just) as, (just) like, and, similar to, be similar to, the same as, be alike, not only... but also, to compare to/with, in the same manner

The proof of Theorem 10 is similar to the proof of Theorem 9 because both Theorems are alike.

Example (introduce an example)

For example, for instance, that is, such as, including, namely

The algorithm has many applications, including image processing, data mining, and schedule planning.

Contrast (link to contrary statements)

However, nevertheless, nonetheless, still, although, even though, though, but, yet, despite, in spite of, in contrast, in comparison, while, whereas, on the other hand, on the contrary

The given data is highly noised. Still, our method returns good results.

Do not overuse linking words. A rule of thumb is to use it in about 30% of all sentences. Read your work carefully and find the parts where readability needs to be improved. Avoid using the same words or formulations over and over. Instead see if you can find synonyms which can replace some of the text.

8.1 Exercise

Choose a linking word of each category and construct a fitting example. Use a linking word and statements different from the given examples.